

ROHIT DIWAKAR

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Links – [Website URL](#) | [dev.to Blog](#) | [GitHub](#)

PROFESSIONAL SUMMARY

- I have overall 12 years of industry experience across various verticals and nearly 4 years of cloud experience working with multiple CSPs (AWS, Azure, GCP) in various capacities.
- Designed and built highly scalable and high-performant infrastructure provisioning architectures based on microservices using managed services and serverless technologies.
- Documentation of product architecture and design using standard design templates.
- Good understanding of latest cloud trends and best practices, including Cloud Native, DevOps and Infra-As-Code.
- Mentored team members, defined best practices, guidelines and blueprints for the team in various capacities.
- Experience in a customer-facing capacity, support, or consulting role.
- Strong written and oral communication skills.
- Ability to relocate/ travel as required.

SKILLS

- **Programming Languages** – Python3, Shell, NodeJS, TypeScript.
- **Containerization Tools** – Docker, Kubernetes.
- **Infra As Code** – Terraform, Ansible, ARM Templates, CloudFormation.
- **Observability & Monitoring** – Datadog, CloudWatch, Events, Azure Monitor, Prometheus, Grafana.
- **Frameworks & Libraries** – FastAPI, Express, Flask, React-Redux.
- **CICD** – GitHub Actions, Jenkins, Azure Pipelines.
- **Testing Frameworks** – Pytest, Mocha, Chai, Jest, Jasmine.
- **Database** – DynamoDB, MongoDB, SQL, Google Firebase Realtime DB & CloudStore.
- **OS** – SLES 12, SLES 15, CentOS, Ubuntu 18, 20, Mac, Windows.

WORK EXPERIENCE

Teradata India Pvt Ltd | Staff Cloud Engineer | 07/2021 – Present

Project – Multi Cloud Provisioning Automation Team (Teradata Vantage IntelliCloud Team)

The scope of my work was to manage and improve multi-cloud site provisioning processes for Vantage service using large DB nodes and its infrastructure. We had to factor in high availability, security, backup, monitoring, etc.

- Designed and deployed cloud agnostic solution for multi-cloud site provisioning. It helped reduce number of services from 42 (for AWS) and 17 (Azure) to only 4 core services that helped improve project maintainability.
- I worked on 4 core REST-based microservices mostly written in Python3 and FastAPI. The services are hosted on AWS account control plane and each service is fronted by API gateway that talks to Step Functions, Lambdas or ECS Task Runners internally. They further integrate with Secret Manager, Ping server, Service Now, DynamoDB and S3 buckets for their functioning. Route53 was used for FQDNs.
- Configured Ansible roles and Terraform modules for infrastructure, component installation (Nessus, Centrify, NXlog) and post-install configurations. Ansible playbooks were also used to upload and run custom scripts on deployed sites.
- Designed CFT and ARM templates to provision various components – Load Balancers, VNet/VPC, subnets, routes, firewall rules, API gateways, S3 buckets, ECS Task Runners. Used parameter files to pass env based variables and config.
- Troubleshooting of failing site provisioning steps. This involved debugging CloudWatch, Azure Monitor logs for each service (mostly Azure Functions, Lambdas and Step Functions) for each provisioning step in the workflow. I also had to debug various K8s clusters to resolve issues. Cloud CLIs (gcloud, aws CLI and az CLI) were heavily used.
- To manage secrets, keys in Secret Managers and Key Vaults, and to define and assign custom roles and policies to service accounts so they can be used to run provisioning steps. Principle of least privilege was used.
- Configured observability, monitoring with Datadog integrations and setup alerts and logging using CloudWatch Events.
- Set up provisioning-host VMs using CloudFormation (CFT) which were used to configure additional tools on deployed sites. VMs are configured on launch by installing tools from Artifactory under UserData section of CFT.
- Created (Azure) Image Definitions and Image Versions for components from VHDs given by component-owner teams (kept in Azure Storage using SAS tokens). Created new offers and workflow configurations from these images and validate their correctness (versions) from a given BOM list.
- Migrated old (and configured new) artifact from outgoing on-prem Artifactory server to cloud-based JFrog Artifactory. This helped team streamline and secure (a previously ad-hoc and chaotic) artifact deployment process.
- Identified and improved various pain points in provision steps (Azure mostly). Some of these steps are – smart retry if provisioning fails or partially failed (instead of destroying already created resources), reduce ARM deployment step to less than 90mins, run unrelated provisioning steps in parallel to save time, show correct error messages when services (mostly Step Functions) failed in backend, stabilize dev environment for testing, warn if certain step is taking longer than usual.

- Created scripts to auto shutdown test VMs and resources over the weekend. This helped the team bring down running costs from \$80,000 to less than \$3000 every weekend.

Xoriant Solutions Pvt Ltd | Senior Software Engineer| 07/2017 – 03/2021

Project – Encompass Consumer Connect (EllieMae) Team

The scope of my work was to create and redesign high traffic systems for Loan Application (for Customers), Borrower App (for Loan Officers) and Admin Portal (internal backend).

- Re-designed frontend, backend services. Improved code coverage to reduce technical debt and improve modularity using NodeJS, TypeScript, React-Redux.
- Demoed POC to use Docker and K8s clusters to quickly setup dev and testing environments for our project.
- Implemented CI/CD pipeline from scratch using GitHub Actions and Jenkins that helped team develop and test features quickly and get immediate feedback.
- Helped design pipelines to auto-merge PRs using Jenkins, GitHub Actions so that new features, fixes are available for future-release branches. Merge conflicts would be notified in Slack channel, and needed to be merged manually.

Quinstreet Software Pvt Ltd | Senior Software Engineer| 11/2015 – 08/2017

Project – Quinstreet Media Platform Tool

- Redesigning front end for an internal media query tool (QMP) using DurandalJS, KnockoutJS so teams across verticals get insight into their digital marketing performance metrics, such as leads, clicks and conversion ratios.
- Improved coding standards to help team write and maintain better code; separated UI from business logic, removed hardcoded and repeating code and moved functions to reusable modules, created templates with data-binding placeholders to generate dynamic data.
- Troubleshoot misc issues with Tableau integration and backend that was written in Python2, Java and PostgreSQL.
- Optimised various sections of QMP tool such as adding pagination, lazy loading and implemented service workers to intercept network requests and fetch and cache resources.

SNS Technologies (Now ForgeAhead) | Senior Software Engineer| 09/2013 – 10/2015

Project – Medsolis, NoodleTools, CK12 – Senior Software Engineer

- Implemented responsive and smart patient dashboard using BackboneJS (UI) and Python2, Flask, DynamoDB (backend). This helped doctors and patients interact with each other and share important instructions.
- Created smart annotation tool for students to use to highlight and annotate various topics while studying topics.
- Played major role in transitioning team from SVN to GitHub by conducting workshops so teams can collaborate and develop code better and faster.

Tata Consultancy Services | Senior Software Engineer| 12/2009 – 09/2013

Project – SET Campaigns (Microsoft), UPSS Team (Vodafone), Airplay Webpages (Bang & Olufsen) – Software Engineer

Technologies – Vanilla JavaScript, Flash Magic tool

- Managed webpages for premium audio products so users can configure their devices, such as configure wi-fi, update firmware, reset settings, etc.
- Created digital marketing campaigns for various customers to promote their products and services during major events, such as Olympics, Super Bowl, etc.
- To monitor and troubleshoot schedulers for major telecom company and to ensure smooth rollouts to new circles. Created scripts and cron jobs to verify if any files have been missed by scheduler; to automate re-running of schedulers, if any.

EDUCATION

- Bachelor of Engineering, Computer Science, 2009 batch, Amravati University.
- Higher & Senior Secondary Schooling, Computer Science, 2005 batch, Naval Public School, Mumbai (CBSE).

PROJECTS

I work on projects for my learning and productivity. Please see my [website](#), [blog](#) and [GitHub](#) for more.

Project – Home NFS Server ([Blog](#))

- I created an NFS server on a Raspberry Pi so I could download movies or shows from my phone on to the server and then stream them on my Android TV or other devices directly. This helped me save a lot of space on my phone and also allowed me to view content on any device (TV, laptop or phone) that are on my home network.

Project – Smart Contact Me Form ([Blog](#) and [GitHub](#))

- I created a smart contact me form for my website. This helps me connect with anybody who wants to reach out via my website. It sends my contact information to the user on their email and sends me a Slack notification.

CERTIFICATIONS

- Microsoft Certified Azure Administrator Associate AZ-104 (Feb 2021)
- Microsoft Certified DevOps Engineer Expert AZ-400 (Feb 2021)